Abstract Algebra II - Homework 1

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Consider the evaluation map defined by substituting x with u:

$$\varphi : K[x] \to L,\ \ g(x) \mapsto g(u)$$

The image of this homomorphism, $\mathrm{Im}(\varphi)$, is $K[u]$ by definition. Because u is algebraic over K, there exists at least one non-zero polynomial in $K[x]$ having u as a root. Hence, $\ker(\varphi) \neq \{0\}$.

Since K is a field, thus $K[x]$ is a PID. Let the monic generator of $\ker(\varphi)$ be $f(x)$ (We can do so because multiplying f by a constant, the ideal generated is unchanged), i.e., $\ker(\varphi) = (f(x))$. By the first isomorphism for rings, we have:

$$K[x]/(f(x)) \cong K[u]$$

1.1 (b)

Since $K[u]$ is a subring of L, and L is an integral domain, so $K[u]$ is also a integral domain. Since $K[x]/(f(x))$ is an integral domain by the isomorphism, $(f(x))$ is a prime ideal, and every non-zero prime ideal in a PID is maximal.

Because $(f(x))$ is maximal, its generator $f(x)$ is irreducible, and furthermore, $K[x]/(f(x))$ is a field.

Finally, $f(x)$ satisfies two properties. Since $f \in \ker\varphi$, we have $f(u) = 0$. And for any $g \in K[x]$ such that $g(u) = 0$, it means that $g \in \ker\varphi = (f(x))$, hence $g(x)$ is divisible by $f(x)$. The converse is also true, if $g(x)$ is divisible by $f(x)$, it means $g(x) \in (f(x)) = \ker(\varphi)$, so $g(u) = 0$.

1.2 (a)

Since $K(u)$ contains all element of the forms $\frac{g(u)}{1}$ where $g(x) \in K[x]$, $K[u]$ automatically lies in $K(u)$, so we have $K[u] \subseteq K(u)$.

But from (b) we know $K[u]$ is a field containing both K and u, and $K(u)$ is the smallest subfield of L meets this requirement. Hence, $K[u] = K(u)$, thus $K[u] \cong K(u)$.

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Since u is algebraic over K, then there exists a unique monic irreducible polynomial $f \in K[x]$ such that $f(u) = 0$. Let the isomorphism from $K(u)$ to $K(u')$ be φ, let $f(x) = a_0 + a_1x + a_2x^2 + \ldots + a_nx^n$.

$$f(u) = a_0 + a_1u + a_2u^2 + \ldots + a_nu^n \in K(u)$$

If we map $f(u)$ to $K(u')$ using φ, we get:

$$\begin{aligned}
\varphi(f(u)) &= \varphi(a_0 + a_1 u + a_2 u^2 + \ldots + a_n x^n) \\
&= \varphi(a_0) + \varphi(a_1)\varphi(u) + \varphi(a_2)\varphi(u^2) + \ldots + \varphi(a_n)\varphi(u^n) \\
&= a_0 + a_1 u' + a_2 u'^2 + \ldots + a_n u'^n \\
&= f(u') = 0
\end{aligned}$$

Hence, u and u' are roots of the same irreducible polynomial $f \in K[x]$.